

HIGH PERFORMANCE COMPUTING Lab Ass-4

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Batch-11

Lab: OpenMP Parallel Loops (From Serial to Parallel)

Task 1: OpenMP Parallel Loops (From Serial to Parallel)

Objective: Confirm OpenMP-ready system.

Steps:

Lscpu

nproc

gcc --version

Expected Observation:

Multiple CPU cores available

GCC installed with OpenMP support

```
lscpu@lscpu:~$ sudo apt update
[sudo] password for lscpu:
get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
get:2 http://archive.ubuntu.com/ubuntu noble InRelease [126 kB]
get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1411 kB]
get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
get:6 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [230 kB]
get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
get:8 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [584u B]
get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [299 kB]
get:11 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [212 kB]
get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.3 kB]
get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [19.9 kB]
get:14 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2369 kB]
get:15 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [548 kB]
get:16 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
get:17 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [516 B]
get:18 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [20.8 kB]
get:19 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [6492 B]
get:20 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
get:21 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [396 B]
get:22 http://archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
get:23 http://archive.ubuntu.com/ubuntu noble/universe amd64 Components [3873 kB]
get:24 http://archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [381 kB]
get:25 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [268 kB]
get:26 http://archive.ubuntu.com/ubuntu noble/multiverse Translation-en [119 kB]
get:27 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
get:28 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8228 B]
get:29 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1723 kB]
get:30 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [222 kB]
get:31 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [179 kB]
get:32 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.4 kB]
get:33 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1328 kB]
get:34 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [311 kB]
get:35 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
get:36 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [13.9 kB]
get:37 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2547 kB]
get:38 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [483 kB]
get:39 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
get:40 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 c-n-f Metadata [556 B]
get:41 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [22.4 kB]
get:42 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [6816 B]
get:43 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
get:44 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [106 B]
get:45 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [48.4 kB]
get:46 http://archive.ubuntu.com/ubuntu noble-backports/main Translation-en [9288 B]
get:47 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7389 B]
get:48 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [368 B]
get:49 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [29.5 kB]
get:50 http://archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [17.9 kB]
get:51 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.8 kB]
get:52 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1404 B]
get:53 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
get:54 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 c-n-f Metadata [116 B]
get:55 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
get:56 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 c-n-f Metadata [116 B]
etched 39.6 MB in 26s (1522 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
126 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Task 2: Serial Baseline – Loop Computation

Objective: Establish serial execution time.

File: sum_serial.c

#include <stdio.h>;

#include <time.h>;

04-02-2026 EOD

```
#define N 100000000
```

```
int main() {  
    double sum = 0.0;  
    clock_t start = clock();
```

```
    for (long i = 0; i < N; i++) {  
        sum += i * 0.5;  
    }
```

```
    clock_t end = clock();  
    double time_spent = (double)(end - start) /  
        CLOCKS_PER_SEC;
```

```
    printf("Serial Sum: %f\n", sum);  
    printf("Execution Time: %f seconds\n", time_spent);  
    return 0;  
}
```

Compile & Run:

```
> gcc sum_serial.c -o sum_serial
```

```
> ./sum_serial
```

Submission: Screenshot showing execution time.

```
GNU nano 7.2 sum_serial.c +
#include <stdio.h>
#include <time.h>

#define N 100000000

int main() {
    double sum = 0.0;
    clock_t start = clock();

    for (long i = 0; i < N; i++) {
        sum += i * 0.5;
    }

    clock_t end = clock();
    double time_spent = (double)(end - start) / CLOCKS_PER_SEC;

    printf("Serial Sum: %f\n", sum);
    printf("Execution Time: %f seconds\n", time_spent);

    return 0;
}

nithish@LAPTOP-A2FOJ865:~$ gcc (Ubuntu xx.x.x) xx.x.x
bash: syntax error near unexpected token `Ubuntu'
nithish@LAPTOP-A2FOJ865:~$ gcc sum_serial.c -o sum_serial
nithish@LAPTOP-A2FOJ865:~$ ./sum_serial
Serial Sum: 2499999975000000.000000
Execution Time: 0.462861 seconds
nithish@LAPTOP-A2FOJ865:~$ |
```

Task 3: OpenMP Parallel Loop

Objective: Parallelize the loop using OpenMP.

File: sum_openmp.c

04-02-2026 EOD

```
#include <stdio.h>;
```

```
#include <omp.h>;
```

```
#define N 100000000
```

```
int main() {
    double sum = 0.0;
    double start = omp_get_wtime();
```

```
#pragma omp parallel for reduction(+:sum)
for (long i = 0; i < N; i++) {
    sum += i * 0.5;
```

```
}
```

```
double end = omp_get_wtime();
```

```
printf(&quot;Parallel Sum: %f\n&quot;, sum);
```

```
printf(&quot;Execution Time: %f seconds\n&quot;, end - start);
```

```
printf(&quot;Threads Used: %d\n&quot;, omp_get_max_threads());
```

```
return 0;
```

```
}
```

Compile:

> gcc -fopenmp sum_openmp.c -o sum_openmp



```
GNU nano 2.9.2 sum_openmp.c
#include <stdio.h>
#include <omp.h>

#define N 100000000

int main() {
    double sum = 0.0;
    double start = omp_get_wtime();

    #pragma omp parallel for reduction(+:sum)
    for (long i = 0; i < N; i++) {
        sum += i * 0.5;
    }

    double end = omp_get_wtime();

    printf("Parallel Sum: %f\n", sum);
    printf("Execution Time: %f seconds\n", end - start);
    printf("Threads Used: %d\n", omp_get_max_threads());

    return 0;
}
```

```
nithish@LAPTOP-A2F0J865:~$ nano sum_openmp.c
nithish@LAPTOP-A2F0J865:~$ gcc -fopenmp sum_openmp.c -o sum_openmp
nithish@LAPTOP-A2F0J865:~$ ./sum_openmp
Parallel Sum: 24999999975000000.000000
Execution Time: 0.146090 seconds
Threads Used: 8
nithish@LAPTOP-A2F0J865:~$ |
```

Task 4: Thread Scaling Experiment

Run with different thread counts:

```
export OMP_NUM_THREADS=1
```

```
./sum_openmp
```

```
export OMP_NUM_THREADS=2
./sum_openmp
```

```
export OMP_NUM_THREADS=4
./sum_openmp
```

```
export OMP_NUM_THREADS=8
./sum_openmp
```

Record the readings in a below table:

```
nithish@LAPTOP-A2FOJ865:~$ export OMP_NUM_THREADS=1
./sum_openmp
Parallel Sum: 2499999975000000.000000
Execution Time: 0.551355 seconds
Threads Used: 1
nithish@LAPTOP-A2FOJ865:~$ |
```

```
nithish@LAPTOP-A2FOJ865:~$ export OMP_NUM_THREADS=2
./sum_openmp
Parallel Sum: 2499999975000000.000000
Execution Time: 0.236198 seconds
Threads Used: 2
nithish@LAPTOP-A2FOJ865:~$ |
```

```
nithish@LAPTOP-A2FOJ865:~$ export OMP_NUM_THREADS=4
./sum_openmp
Parallel Sum: 2499999975000000.000000
Execution Time: 0.143396 seconds
Threads Used: 4
nithish@LAPTOP-A2FOJ865:~$ |
```

```
nithish@LAPTOP-A2FOJ865:~$ export OMP_NUM_THREADS=8
./sum_openmp
Parallel Sum: 2499999975000000.000000
Execution Time: 0.117494 seconds
Threads Used: 8
nithish@LAPTOP-A2FOJ865:~$ |
```

Threads	Execution Time (s)
1	0.551355 seconds
2	0.236198 seconds

4	0.143396 seconds
8	0.117494 seconds

Task 5:
Speedup

$$S(p) = T_1 / T_p$$

Efficiency

$$E(p) = S(p) / p$$

Compute for 2, 4, 8 threads.

Submission: Table with Speedup & Efficiency.

Step 1: Identify execution times

From **Task 4**, you already have:

- **T1** → execution time with **1 thread**
- **T2** → execution time with **2 threads**
- **T4** → execution time with **4 threads**
- **T8** → execution time with **8 threads**

Step 2: Use formulas

Speedup:

$$S(p) = T_1 / T_p \quad S(p) = T_1 / T_p$$

Efficiency:

$$E(p) = S(p) / p \quad E(p) = S(p) / p$$

Step 3: Example calculation

Assume (example values):

- $T1 = 1.20 \text{ s}$
- $T2 = 0.65 \text{ s}$
- $T4 = 0.38 \text{ s}$
- $T8 = 0.30 \text{ s}$

For 2 threads:

- $S(2) = 1.20 / 0.65 = \mathbf{1.84}$
- $E(2) = 1.84 / 2 = \mathbf{0.92}$

For 4 threads:

- $S(4) = 1.20 / 0.38 = \mathbf{3.15}$
- $E(4) = 3.15 / 4 = \mathbf{0.79}$

For 8 threads:

- $S(8) = 1.20 / 0.30 = \mathbf{4.00}$
- $E(8) = 4.00 / 8 = \mathbf{0.50}$

Threads (p)	Time T_p (s)	Speedup $S(p)$	Efficiency $E(p)$
2	$T2$	$T1 / T2$	$S(2) / 2$
4	$T4$	$T1 / T4$	$S(4) / 4$
8	$T8$	$T1 / T8$	$S(8) / 8$

1. Why does OpenMP work well for this loop?

OpenMP works well here because every loop iteration is independent. Each thread can calculate its part without waiting for others, and at the end OpenMP safely adds all the partial results together using the reduction clause.

2. What happens if loop iterations are dependent?

If the iterations depend on each other, running them in parallel can give wrong results. This happens because multiple threads may try to read or update the same data at the same time, causing race conditions. Adding synchronization can fix this, but it also slows down the program.

3. Why does speedup stop improving after some threads?

After a certain number of threads, performance stops improving because the system has limited CPU cores. Also, managing many threads takes extra time, and they start competing for memory. Some parts of the program still run serially, which limits overall speedup.

4. Is OpenMP suitable for multi-node systems? Why or why not?

No, OpenMP is mainly meant for systems where all threads share the same memory, like a single machine. In multi-node systems, each node has its own memory, so communication is needed. For that, MPI is more suitable than OpenMP.